Paolo Joseph Baioni

HPC & Scientific Computing Engineer

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Professional Summary

HPC and GPU software engineer specialising in modern C++ (C++20/23), CUDA/ROCm, and high-performance scientific applications.
Experience in performance-portable software across CPU/GPU architectures and hands-on work with Linux-based HPC systems, including cluster design, deployment, and distributed workload, as well as in mentoring and technical supervision within HPC environments.
Background in numerical methods and computational physics, with contributions to peer-reviewed research and in-house research and development software.

Education

11/2021 – 02/2025 **PhD in Mathematical Models and Methods in Engineering**, *Laboratory for Modelling and Scientific Computing (MOX)*, Mathematics Department, Politecnico di Milano
Design and development of C++20 code for GPUs and CPUs (Thrust, std::execution/TBB, CUDA, HIP).
PhD Thesis: *"Material Point Method for Compressible Flows: a Portable, Massively Parallel Implementation for High-Performance Computing Architectures."*
Supervisor: prof. Carlo de Falco, carlo.defalco@polimi.it

2021 **National Habilitation**, Senior Nuclear Engineer

10/2020 – 10/2021 **MSc in Nuclear Eng.**, Nuclear System Physics, *Politecnico di Milano*, 107/110

09/2017 – 10/2020 **MSc in Math. Eng.**, Computational Sciences and Engineering, *Politecnico di Milano*, 110/110

09/2013 – 07/2017 **BSc in Ingegneria Matematica**, *Politecnico di Milano*, 110/110

Professional Experience

12/2023 – present (Permanent Position) **Technical Manager - MOX Laboratory for Modelling and Scientific Computing**, Mathematics Department, Politecnico di Milano

- Development and optimisation of scientific software from scratch using modern C++ and GPU technologies (CUDA, ROCm), with a focus on performance portability
- Developed software later adopted and extended in subsequent research activities
- Analysis and extension of existing codebases, including identification of reusable patterns and adaptation to GPUs
- Design, deployment, and lifecycle management of on-premise Linux-based HPC clusters, including system integration and architecture definition for multi-node environments
- Configuration and operation of HPC software stacks and workload managers (OpenPBS, SLURM)
- Adaptation of HPC software stacks to heterogeneous Linux clusters, including system-level debugging and compatibility patches
- Collaboration with academic and industry partners on scientific computing projects, contributing to code optimisation and scalable software design
- Principal Investigator (PI) of Department projects at CINECA (G100, Leonardo), enabling access to national HPC resources and supporting advanced research activities
- Leadership of an HPC infrastructure renewal initiative, securing funding for GPU-enabled systems and high-speed interconnects
- Supervision of technical staff for cluster operations and Linux infrastructure
- Co-supervision of MSc and PhD students on HPC, GPU computing, and parallel programming

2022 – present **Teaching Assistant - Scientific Computing MSc Courses**, Politecnico di Milano

- Deliver scientific computing labs at MSc level — see e.g.: github.com/pacs-course/pacs-Labs/tree/main/Labs, github.com/HPC-Courses/AMSC-Labs/tree/main/Labs/2025-26
- Cover C++20/23, MPI, GPU programming, build systems, debugging and performance tools, HPC software stacks and abstraction layers, numerical methods and mathematical modelling
- Mentor graduate students in scientific computing and HPC topics

Key Technical Skills

Programming	C++20/23, CUDA-C, HIP/ROCm, MPI, C, Eigen, Thrust, std::execution/TBB, pybind11, Make, shell, Python (Numba, scripting)
Software & Tools	git/GitHub, Podman, Apptainer, Spack, CMake, Doxygen, PBS/SLURM, gdb, lcov, valgrind, Nsight
Numerical Methods	FEM, DG, PIC/MPM, Numerical Linear Algebra, Symplectic Integrators, Runge-Kutta methods
Systems	Linux (Debian, Arch, RHEL), HPC cluster deployment, virtualisation tools (iDRAC, iLO, xClarity)
Scientific Software	○ SOLPS-ITER, coupled Finite Volumes and Monte Carlo code for the simulation of boundary plasma in magnetic confinement fusion devices, including B2.5 (plasma) and EIRENE (WCU MC) ○ PyFR, solver for advection–diffusion using Flux Reconstruction, designed for high-performance and portability across CPU/GPU architectures via a Python-based domain-specific templating layer

Professional Development

Red Hat & Coursera	RHEL, Ansible, Kubernetes, Amazon Simple Storage Service (S3) (2025-2026)
NVIDIA Courses	Scaling Workloads Across Multiple GPUs with CUDA C++, GPU Acceleration with the C++ Standard Library, Building Container Images for HPC, Accelerated Computing with CUDA Python, Accelerating CUDA C Applications with Concurrent Streams, and additional courses (2024–2025)
CINECA Courses	Build Systems and Package Managers in HPC, Parallel Computing on Traditional Core-Based and Emerging GPU-Based Architectures, Numerical Methods for Parallel CFD (2021-2023)
Conferences and Workshops	Continuous learning through attendance at various industry workshops, EuroHPC hackathons, and scientific conferences on HPC, AI, Scientific Machine Learning, Quantum Computing

Publications and Conferences

2025	**Workshop Poster (selected)**, *Portable Parallel Computing: C++ Strategies for Cross-Architecture Performance*, Baioni P.J., Benacchio T., Capone L., de Falco C., E4: AI, HPC & QUANTUM 2025
2024	**Journal Paper**, *Portable, massively parallel implementation of a material point method for compressible flows*, Baioni P.J., Benacchio T., Capone L., de Falco C., Computational Particle Mechanics, Springer. DOI: 10.1007/s40571-024-00864-2, full text https://rdcu.be/d20u2
2024	**Review activity**, Reviewer for the Journal of Computational Particle Mechanics, Springer
2023	**Conference Proceedings**, *GPUs Based Material Point Method for Compressible Flows*, Baioni P.J., Benacchio T., Capone L., de Falco C. VIII International Conference on Particle-Based Methods, DOI: 10.23967/c.particles.2023.026
2023	**Conference Proceedings**, *MS08 - Particle-Based Methods In Applied And Industrial Sciences*, Baioni P.J., Crescenzio N., Fois M., Moreno Martinez L. The XVI biennial congress of SIMAI, link: https://web.unibas.it/simai2023/BookOfAbstracts.pdf
2023	**Conference Proceedings**, *Material Point Method for Compressible CFD on GPUs*, Baioni P.J., Benacchio T., Capone L., de Falco C., Pellegrino E. The XVI biannual congress of SIMAI

FOSS Software

2020	Development of a C++14 code for a numerical methods and programming university course project, implementing a Deep Neural Network (DNN) from scratch, relying on the C++ Standard Library and Eigen; code available at https://github.com/pjbaioni/neural-net

Professional Memberships

2023 – present	Member of INdAM, National Scientific Computing Group (GNCS)
2023 – present	Member of Italian Society for Applied and Industrial Mathematics (SIMAI)
2022 – present	Enrolled in the Order of Engineers of the Metropolitan City of Milan

Languages

Italian	Native speaker
English	C1 (TOEIC 2016); teaching MSc courses in English since 2022